

MINI PROJECT REPORT FILE



HOTEL MANAGEMENT SYSTEM

Submitted by -

1. Aditya Yadav
2. Himanshu Singh Rajput
3. Ansh Dubey
4. Satwik Vats
5. Tejendra Singh Chauhan

Submitted to -

Mr. Vikas Kumar

DECLARATION

We hereby declare that the project report entitled "Hotel Management System" submitted by us to GLA UNIVERSITY is a bona fide work completed by us under the guidance of Mr. Vikas Kumar, Department of Computer Engineering

We further declare that the project has been prepared entirely by us and that no part of it has been copied from any other project or published sources. The work is original and is not a reproduction of any previous work, except where the information is stated to be derived from published sources. We have acknowledged all sources of information and assistance in the project report. This work has not been submitted elsewhere for the award of any degree or diploma or for any other purpose.

Members	Name	Univ. Roll Number
Member 1	Aditya Yadav	2215000132
Member 2	Himanshu Singh Rajput	2215000809
Member 3	Ansh Dubey	2215000280
Member 4	Satwik Vats	2215001602
Member 5	Tejendra Singh Chauhan	2215001856

Acknowledgement

We are profoundly grateful for the guidance, support, and encouragement we received throughout the development of our Hotel Management Project.

First and foremost, we extend our heartfelt thanks to our project guide, [Mr. Vikas Kumar], for their valuable advice, constructive feedback, and constant motivation during every phase of this project. Their insights and expertise played a crucial role in helping us achieve our objectives effectively.

We also express our sincere gratitude to [GLA University] for providing us with the necessary resources, infrastructure, and an environment conducive to learning and innovation.

Additionally, we are thankful to our peers and friends who offered their suggestions and feedback, which helped us refine our project.

Lastly, we would like to acknowledge the unwavering support of our families for their encouragement and understanding throughout this journey.

This project has been a remarkable learning experience, allowing us to apply our skills practically and explore the fascinating domain of travel website development.

Certificate

This is to certify that

Aditya Yadav, Himanshu Singh Rajput, Ansh Dubey, Satwik Vats, Tejendra Singh Chauhan

has successfully completed the project titled

"Hotel Management System"

as part of the requirements for B. Tech course. The project has been carried out under the guidance of **Mr. Vikas Kumar** has met the standards expected by

GLA UNIVERSITY

The project demonstrates significant effort, dedication, and practical application of knowledge in web development and has contributed to gaining a deeper understanding of full-stack development principles.

INDEX

S.NO: -	Topic
1	Introduction, About the Project
2	The Main Objective of the Project
3	Frontend and Backend Software used
4	Listing Out testing technology-
5	Working code and Screenshot
6	Module Description
7	Hardware and software requirement
8	DFD (0 level, 1 level and 2 level)
9	Future Scope
10	Bibliography and References

Introduction of the project:

Hotel management system project is a software solution designed to automate and streamline the operations of a hotel, including reservation management, room assignment, billing, inventory management, and guest services. The system aims to provide an efficient and cost-effective way for hotels to manage their day-to-day activities and improve the guest experience.

The hotel management system typically includes a front-end interface for hotel staff to manage reservations, check-ins, and checkouts, as well as a back-end database to store guest information and track inventory. The system may also integrate with other software solutions, such as accounting and customer relationship management (CRM) software.

Some of the key features of a hotel management system include online booking, room availability tracking, room assignment automation, billing and invoicing, inventory management, housekeeping and maintenance management, and guest feedback management.

The benefits of using a hotel management system include increased efficiency, improved accuracy, reduced administrative costs, enhanced guest experience, and improved revenue management. With the help of a hotel management P a g e | 2 system, hotels can streamline their operations, improve guest satisfaction, and increase profitability

Objective of the project:

The main objective of a hotel management system project is to provide an efficient and reliable way to manage all the activities of a hotel. This system aims to automate and streamline the processes involved in hotel operations such as room booking, customer check-in and check-out, inventory management, staff management, and billing.

The hotel management system project can provide several benefits such as:

Improved guest experience: By automating the booking and check-in process, guests can have a smoother and hassle-free experience.

Increased efficiency: With automation, the hotel staff can save time on routine tasks, which can improve their productivity and reduce errors.

Better inventory management: The system can help manage inventory, such as room availability, food and beverage, and other hotel supplies, which can reduce waste and optimize costs.

Accurate billing and reporting: The system can generate accurate bills and financial reports, which can help in better financial management.

Overall, the main objective of the hotel management system project is to enhance the guest experience, optimize hotel operations, and increase profitability.

Project category:

Hotel management system project using php language and my SQL database connectivity project category

The Hotel Management System project is a web-based application that manages the day-to-day activities of a hotel. It involves managing hotel bookings, room allocation, guest check-in, and check-out, billing, and other administrative tasks. The system is built using PHP programming language and MySQL database connectivity.

Project Category: Web Application

Project Features:

User authentication and authorization system

Dashboard for the hotel staff to manage bookings, room allocation, and guest details

Online reservation system for customers to book rooms and view available rooms

Payment gateway integration for secure online payments

Room availability and pricing management

Automated check-in and check-out process

Billing and invoicing system

Guest feedback system to improve the hotel's services

Reporting and analytics to track the hotel's performance.

System and Hardware Requirements

Technologies Used:

PHP programming language

MySQL database connectivity

HTML, CSS, and JavaScript for front-end development

Bootstrap framework for responsive web design

Apache web server for hosting the application

Payment gateway integration APIs

Project Deliverables:

Source code of the Hotel Management System web application

Database schema and sample data

User manual and installation guide

Project documentation including design, testing, and deployment details

Project Duration: Depending on the complexity of the features,

Working code with Screenshots

Nav bar code



[Home](#) [About](#) [Services](#) [Explore](#) [Contact](#)

[Login](#) [Sign Up](#)

Simple - Unique - Friendly

Make Yourself At Home
In Our **Hotel.**





Any Room Type



Search



ABOUT US

The Best Holidays Start Here!

With a focus on quality accommodations, personalized experiences, and seamless booking, our platform is dedicated to ensuring that every traveler embarks on their dream holiday with confidence and excitement.

Read More

OUR LIVING ROOM

The Most Memorable Rest Time Starts Here.



Skyline Premium Room

Compact and cozy room ideal for solo or couple stays with modern amenities.

Price: ₹2004

Guests: 2

Snahat: Loved the decor! (3★)

Ishtita: Could improve the lighting. (4★)

Book Now



Serenity Standard Room

Compact and cozy room ideal for solo or couple stays with modern amenities.

Price: ₹4506

Guests: 2

Sana: Amazing view and experience. (5★)

Mansaj: Perfect stay! (3★)

Book Now



Amber Woods Room

Compact and cozy room ideal for solo or couple stays with modern amenities.

Price: ₹7562

Guests: 2

Sibya: Loved the decor! (5★)

Rishav: Perfect stay! (5★)

Book Now



Palm Grove Executive

Compact and cozy room ideal for solo or couple stays with modern amenities.

Price: ₹1662

Guests: 2

Ramya: Perfect stay! (4★)

Ishtita: Loved the decor! (3★)

Book Now



Comfort Stay Executive

Compact and cozy room ideal for solo or couple stays with modern amenities.

Price: ₹9648

Guests: 2

Pritya: Very clean and comfortable. (5★)

Mansaj: Amazing view and experience. (4★)

Book Now



Golden Horizon Room

Spacious room for two with elegant furnishings and all essentials.

Price: ₹5500

Guests: 2

Karan: Perfect stay! (5★)

Ravi: Amazing view and experience. (4★)

Book Now

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<link
href="https://cdn.jsdelivr.net/npm/remixicon@4.0.0/fonts/remixicon.css"
rel="stylesheet"
/
<link rel="stylesheet" href="styles.css" />
<title>ROOMBOOKZ</title>
</head>
<body>
<header class="header">
<nav>
<div class="nav_bar">
<div class="logo">
<a href="#"></a>
</div>
<div class="nav_menu_btn" id="menu-btn">
<i class="ri-menu-line"></i>
</div>
</div>
<div>
<ul class="nav_links" id="nav-links">
<li><a href="#home">Home</a></li>
<li><a href="#about">About</a></li>
<li><a href="#service">Services</a></li>
<li><a href="#explore">Explore</a></li>
<li><a href="#contact">Contact</a></li>
</ul>
<a href="login.html" class="btn nav_btn" id="nav-login-btn">Login</a>
<a href="signup.html" class="btn nav_btn" id="nav-signup-btn">Sign Up</a>
</div>
</div>
<div id="profile-dropdown" class="profile-dropdown" style="display:none; align-items:center; gap:10px;">
<a href="profile.html" id="profile-avatar-link" style="display:inline-flex; align-items:center;">

</a>
<span id="profile-name"></span>
<a href="profile.html" class="btn" id="profile-btn">Profile</a>
<button id="logout-btn">Logout</button>
</div>
</nav>
<div class="section__container header__container" id="home">
<p>Simple - Unique - Friendly</p>
<h1>Make Yourself At Home<br />In Our <span>Hotel</span>.</h1>
</div>
</header>

<section class="section__container search__container">
<form id="search-form" class="search-bar">
<input type="text" name="location" placeholder="Location" />
<input type="date" name="checkin" required />
<input type="date" name="checkout" required />
<input type="number" name="guests" min="1" placeholder="Guests" required />
<select name="roomType">
<option value="">Any Room Type</option>
<option value="single">Single</option>
<option value="double">Double</option>
<option value="suite">Suite</option>
</select>
<button type="submit" class="btn">Search</button>
</form>
</section>

<section class="section__container about__container" id="about">
<div class="about_image">

</div>
<div class="about_content">
<p class="section__subheader">ABOUT US</p>
<h2 class="section__header">The Best Holidays Start Here</h2>
<p class="section__description">
With a focus on quality accommodations, personalized experiences, and seamless booking, our platform is dedicated to ensuring that every traveler embarks on their dream holiday with confidence and excitement.
</p>
<div class="about_btn">
<button class="btn">Read More</button>
</div>
</div>
</section>

<section class="section__container room__container">
<p class="section__subheader">OUR LIVING ROOM</p>
<h2 class="section__header">The Most Memorable Best Time Starts Here.</h2>
<div class="room_grid" id="room-list"></div>
</div>
</section>

<section class="service" id="service">
<div class="section__container service__container">
<div class="service__content">
<p class="section__subheader">SERVICES</p>
<h2 class="section__header">Strive Only For The Best.</h2>
<ul class="service_list">
<li>
<span><i class="ri-shield-star-line"></i></span>
High Class Security
</li>
<li>
<span><i class="ri-24-hours-line"></i></span>
24 Hours Room Service
</li>
<li>
<span><i class="ri-headphone-line"></i></span>
Conference Room
</li>
<li>
<span><i class="ri-map-2-line"></i></span>
Tourist Guide Support
</li>
</ul>
</div>
</div>
</section>
```

```
</ul>
</div>
</div>
</section>

<section class="section__container banner__container">
<div class="banner_content">
<div class="banner_card">
<h4>25</h4>
<p>Properties Available</p>
</div>
<div class="banner_card">
<h4>350</h4>
<p>Bookings Completed</p>
</div>
<div class="banner_card">
<h4>600</h4>
<p>Happy Customers</p>
</div>
</div>
</section>

<section class="explore" id="explore">
<p class="section__subheader">EXPLORE</p>
<h2 class="section__header">What's New Today.</h2>
<div class="explore_bg" style="position:relative;">
<div class="explore_content" style="position:relative;z-index:1;">
<p class="section__description">10th MAR 2023</p>
<h4>A New Menu Is Available In Our Hotel.</h4>
<button class="btn">Continue</button>
</div>
</div>
</section>

<footer class="footer" id="contact">
<div class="section__container footer__container">
<div class="footer_col">
<div class="logo">
<a href="#home"></a>
</div>
<p class="section__description">
Discover a world of comfort, luxury, and adventure as you explore our curated selection of hotels, making every moment of your getaway truly extraordinary.
</p>
<button class="btn">Book Now</button>
</div>
<div class="footer_col">
<h4>QUICK LINKS</h4>
<ul class="footer_links">
<li><a href="#">Browse Destinations</a></li>
<li><a href="#">Special Offers & Packages</a></li>
<li><a href="#">Room Types & Amenities</a></li>
<li><a href="#">Customer Reviews & Ratings</a></li>
<li><a href="#">Travel Tips & Guides</a></li>
</ul>
</div>
<div class="footer_col">
<h4>OUR SERVICES</h4>
<ul class="footer_links">
<li><a href="#">Concierge Assistance</a></li>
<li><a href="#">Flexible Booking Options</a></li>
<li><a href="#">Airport Transfers</a></li>
<li><a href="#">Wellness & Recreation</a></li>
</ul>
</div>
<div class="footer_col">
<h4>CONTACT US</h4>
<ul class="footer_links">
<li><a href="#">rayalpark@info.com</a></li>
</ul>
<div class="footer_socials">
<a href="#"></a>
<a href="#"></a>
<a href="#"></a>
<a href="#"></a>
</div>
</div>
</div>
<div class="footer_bar">
Copyright © 2023 Web Design Mastery. All rights reserved.
</div>
</footer>

<script src="https://unpkg.com/scrollreveal"></script>
<script src="main.js"></script>
<script src="search.js"></script>
</body>
</html>
```

HOTEL ROOMS CODE

```
const menuBtn = document.getElementById("menu-btn");
const navLinks = document.getElementById("nav-links");
const menuBtnIcon = menuBtn.querySelector("i");

menuBtn.addEventListener("click", () => {
  navLinks.classList.toggle("open");

  const isOpen = navLinks.classList.contains("open");
  menuBtnIcon.setAttribute("class", isOpen ? "ri-close-line" : "ri-menu-line");
});

navLinks.addEventListener("click", () => {
  navLinks.classList.remove("open");
  menuBtnIcon.setAttribute("class", "ri-menu-line");
});

const scrollRevealOption = {
  distance: "50px",
  origin: "bottom",
  duration: 1000,
};

ScrollReveal().reveal(".header__container p", {
  ...scrollRevealOption,
});

ScrollReveal().reveal(".header__container h1", {
  ...scrollRevealOption,
  delay: 500,
});

ScrollReveal().reveal(".about__image img", {
  ...scrollRevealOption,
  origin: "left",
});

ScrollReveal().reveal(".about__content .section__subheader", {
  ...scrollRevealOption,
  delay: 500,
});

ScrollReveal().reveal(".about__content .section__header", {
  ...scrollRevealOption,
  delay: 1000,
});

ScrollReveal().reveal(".about__content .section__description", {
  ...scrollRevealOption,
  delay: 1500,
});

ScrollReveal().reveal(".about__btn", {
  ...scrollRevealOption,
  delay: 2000,
});

ScrollReveal().reveal(".room__card", {
  ...scrollRevealOption,
  interval: 500,
});

ScrollReveal().reveal(".service__list li", {
  ...scrollRevealOption,
  interval: 500,
  origin: "right",
});

const API_BASE = 'http://localhost:5000/api';

const navLoginBtn = document.getElementById('nav-login-btn');
const navSignupBtn = document.getElementById('nav-signup-btn');
const profileDropdown = document.getElementById('profile-dropdown');
const profileName = document.getElementById('profile-name');
const profileBtn = document.getElementById('profile-btn');
const logoutBtn = document.getElementById('logout-btn');

function updateUIOnLogin(user) {
  if (navLoginBtn) navLoginBtn.style.display = 'none';
  if (navSignupBtn) navSignupBtn.style.display = 'none';
  if (profileDropdown) profileDropdown.style.display = 'flex';
  if (profileName) profileName.textContent = user && user.name ? user.name : 'My Profile';
}

function updateUIOnLogout() {
  if (navLoginBtn) navLoginBtn.style.display = 'inline-block';
  if (navSignupBtn) navSignupBtn.style.display = 'inline-block';
  if (profileDropdown) profileDropdown.style.display = 'none';
  if (profileName) profileName.textContent = '';
}

logoutBtn.onclick = function() {
  localStorage.removeItem("token");
  updateUIOnLogout();
};

profileBtn.onclick = function() {
  window.location.href = 'profile.html';
};

const searchForm = document.getElementById('search-form');
if (searchForm) {
  searchForm.onsubmit = function(e) {
    e.preventDefault();
    const params = new URLSearchParams();
    params.set('location', searchForm.location.value);
    params.set('checkin', searchForm.checkin.value);
    params.set('checkout', searchForm.checkout.value);
    params.set('guests', searchForm.guests.value);
    params.set('roomType', searchForm.roomType.value);
    window.location.href = 'search.html?' + params.toString();
  };
}

const bookingForm = document.querySelector('.booking__form');
if (bookingForm) {
  bookingForm.onsubmit = async function(e) {
    e.preventDefault();
    const token = localStorage.getItem("token");
    if (!token) {
      openAuthModal('login');
      return;
    }
    const checkIn = bookingForm.querySelector('input[placeholder="Check In"]').value;
    const checkOut = bookingForm.querySelector('input[placeholder="Check Out"]').value;
    const guest = bookingForm.querySelector('input[placeholder="Guest"]').value;
    const res = await fetch(`${API_BASE}/bookings`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json', Authorization: 'Bearer ' + token },
      body: JSON.stringify({ checkIn, checkOut, guest })
    });
    const data = await res.json();
    let feedback = document.getElementById('booking-feedback');
    if (!feedback) {
      feedback = document.createElement('div');
      feedback.id = 'booking-feedback';
      feedback.className = 'booking-feedback';
      bookingForm.appendChild(feedback);
    }
    if (res.ok) {
      feedback.textContent = 'Booking successful!';
      bookingForm.reset();
    } else {
      feedback.textContent = data.message || 'Booking failed!';
    }
  };
}

const bookNowBtns = document.querySelectorAll('.room__card .btn');
bookNowBtns.forEach(btn => {
  btn.onclick = function(e) {
    e.preventDefault();
    window.scrollTo({ top: document.querySelector('.booking__container').offsetTop - 40, behavior: 'smooth' });
    document.querySelector('.booking__form input[placeholder="Guest"]').focus();
  };
});

window.addEventListener('DOMContentLoaded', async () => {
  const roomList = document.getElementById('room-list');
  if (roomList) {
    fetch('mock-rooms.json')
      .then(res => res.json())
      .then(rooms => {
        roomList.innerHTML = "";
        rooms.filter(room => {
          const removeNames = ['family garden retreat', 'executive cityscape room', 'deluxe ocean view'];
          return !removeNames.includes(room.name.trim().toLowerCase());
        }).slice(0, 6).forEach(room => {
          roomList.innerHTML += `
<div class="room__card">
  <div class="room__card__image">
    
  </div>
  <div class="room__card__details">
    <h4>${room.name}</h4>
    <p>${room.description}</p>
    <p>Price: *${room.price}</p>
    <p>Guests: ${room.guests}</p>
    <div class="room__reviews">
      ${room.reviews.map(r => `<p><b>${r.user}</b> ${r.comment} (${r.rating}★)</p>`).join("")}
    </div>
    <button class="btn">Book Now</button>
  </div>
</div>
`;
        });
      });
  };
});

const token = localStorage.getItem("token");
if (token) {
  try {
    const res = await fetch(`${API_BASE}/users/profile`, {
      headers: { Authorization: 'Bearer ' + token }
    });
    const data = await res.json();
    if (data && data.name) {
      updateUIOnLogin(data);
    } else {
      updateUIOnLogout();
    }
  } catch (e) {
    updateUIOnLogout();
  }
} else {
  updateUIOnLogout();
};
});
```

Available Rooms

Rooms found: 10
Params: {"location": "Santipur, Uttar Pradesh, India", "checkin": "2025-04-07", "checkout": "2025-04-22", "guests": 1, "roomType": ""}



Skyline Premium Room

2 Guests Single Kolkata

First aid kit Toiletries available Queen Sized Bed In house Restaurant
24/7 check in Security

Compact and cozy room ideal for solo or couple stays with modern amenities.

Your name

Your comment
Add Review

Rating: 3.5 (2 reviews)

- S Sneha

Loved the decor!
- I Ishita

Could improve the lighting.

23004 Book Now



Palm Grove Executive

2 Guests Single Kolkata

Dress Hanger Queen Sized Bed In house Restaurant Twin Single Bed
Toiletries available

Compact and cozy room ideal for solo or couple stays with modern amenities.

Your name

Your comment
Add Review

Rating: 3.5 (2 reviews)

- K Karan

Perfect stay!
- I Ishita

Loved the decor!

26262 Book Now



Serenity Standard Room

2 Guests Single Kolkata

TV Modern wardrobe Toiletries available CCTV cameras Power backup
Attached bathroom

Compact and cozy room ideal for solo or couple stays with modern amenities.

Your name

Your comment
Add Review

Rating: 4.0 (2 reviews)

- S Sana

Amazing view and experience.
- M Manoj

Perfect stay!

24306 Book Now



Comfort Stay Executive

2 Guests Single Kolkata

Reception Queen Sized Bed Free WiFi CCTV cameras In house Restaurant

Compact and cozy room ideal for solo or couple stays with modern amenities.

Your name

Your comment
Add Review

Rating: 5.0 (2 reviews)

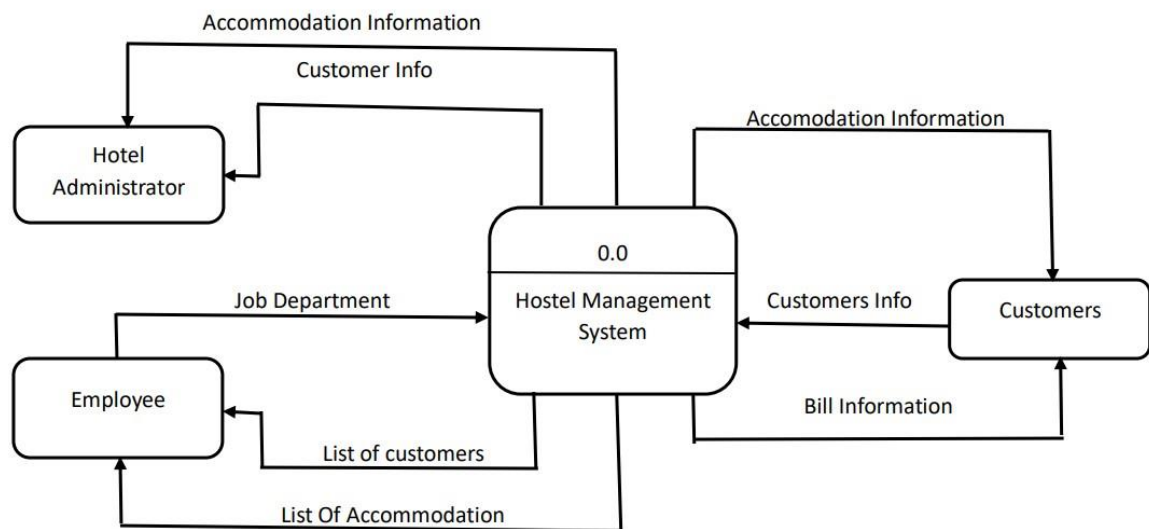
- P Priya

Very clean and comfortable.
- M Manoj

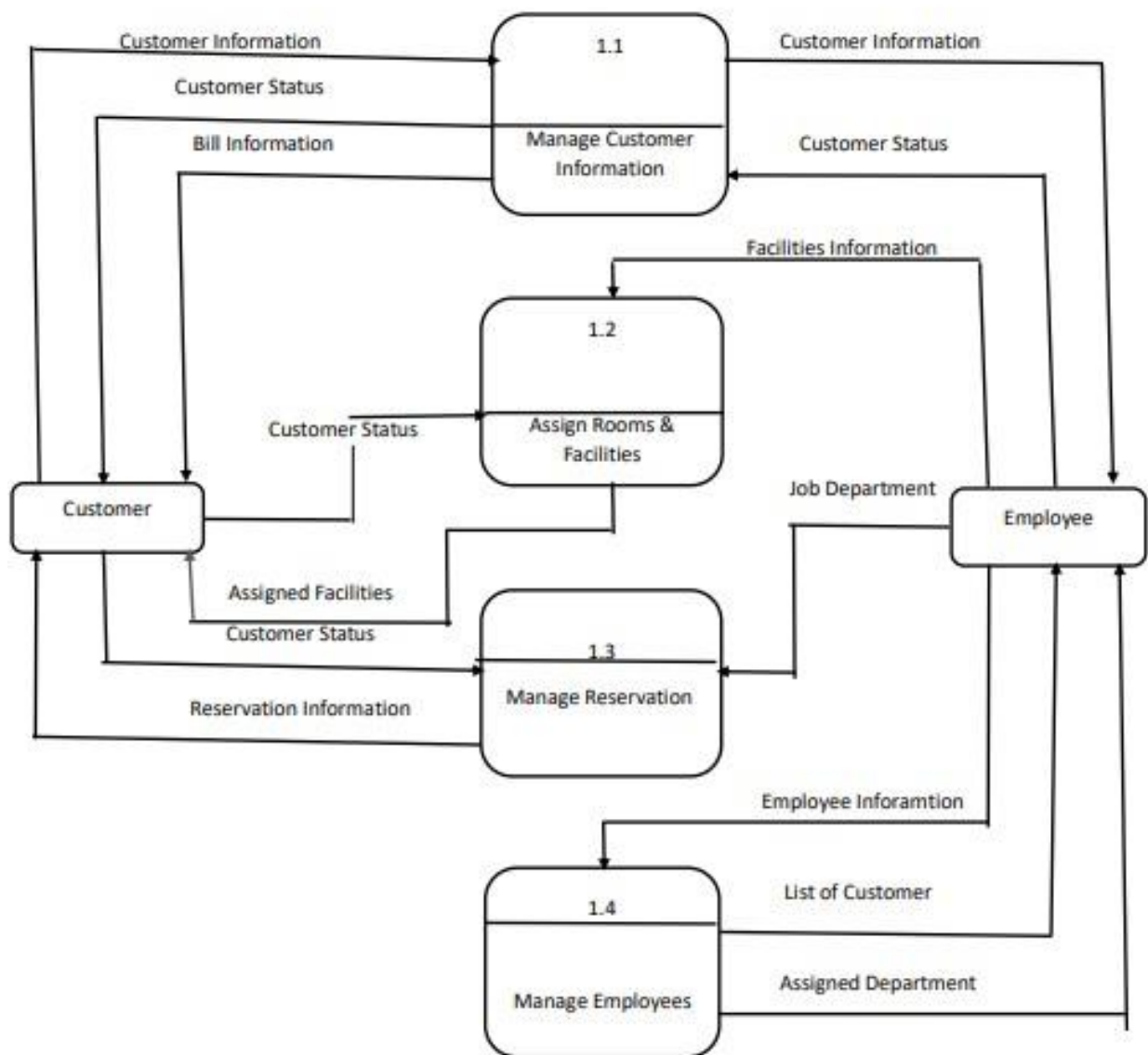
Amazing view and experience.

28668 Book Now

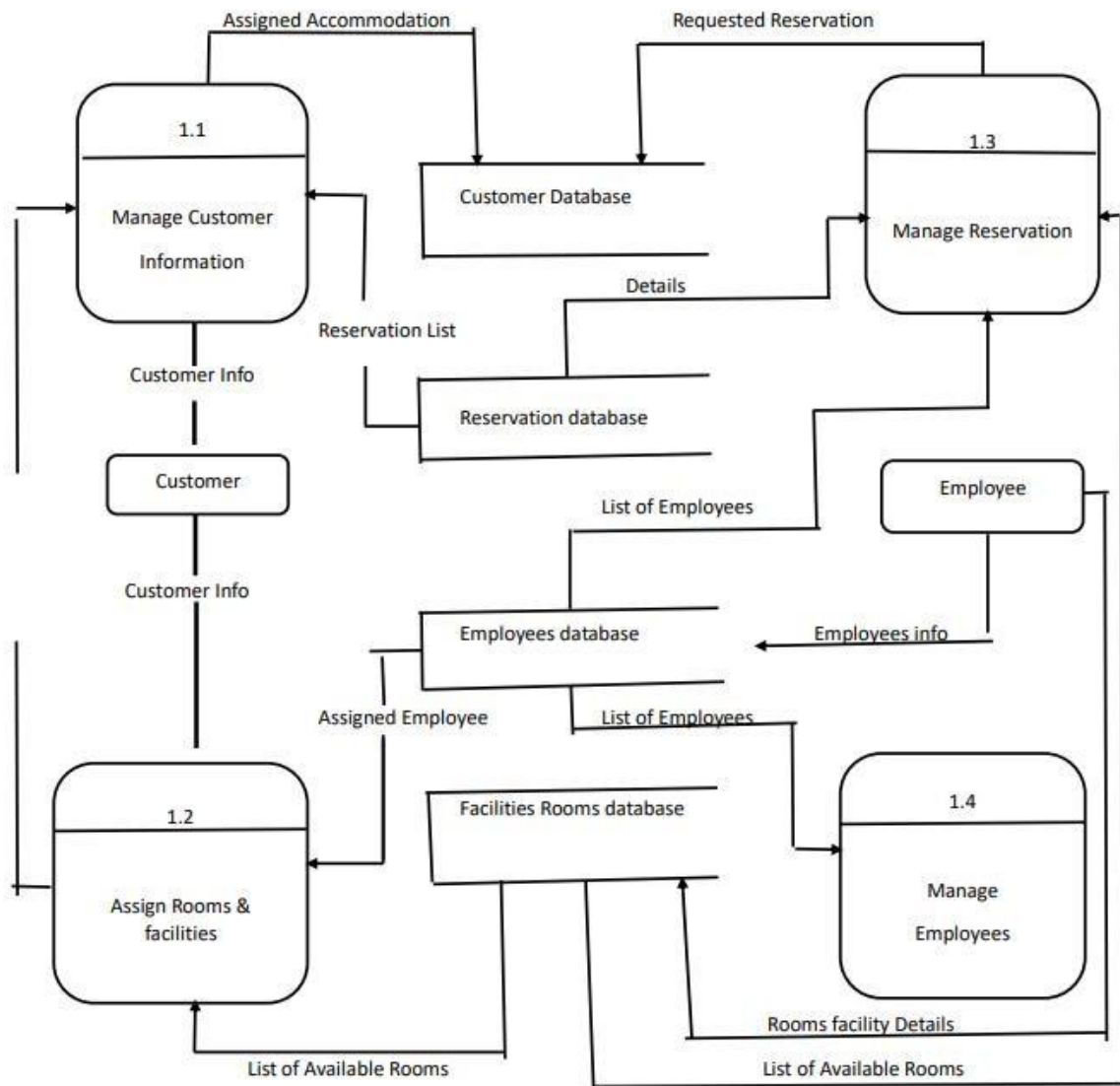
Data Flow Diagram: -



Context Diagram for Hotel Management System



Level 1 DFD for Hotel Management System



Level 2 DFD for Hotel Management System

5. A complete structure which include number of modules:

The Hotel Management System will consist of the following modules:

Login Module: This module will allow users to log in to the system using their username and password.

Room Management Module: This module will allow hotel managers to manage room information such as room type, availability, price, and room amenities.

Reservation Module: This module will allow hotel managers to make room reservations for guests. The module will also allow guests to view room availability, room rates, and make room reservations online.

Guest Management Module: This module will allow hotel managers to manage guest information such as guest name, address, contact details, and booking history.

Billing Module: This module will allow hotel managers to generate bills for guests based on their room reservation and other expenses such as food, laundry, etc.

Data Structures:

The following data structures will be used in the Hotel Management System:

Room Information: Room Number, Room Type, Room Price, Room Availability, Room Amenities.

Guest Information: Guest Name, Guest Address, Guest Contact Details, Booking History.

Reservation Information: Room Number, Check-in Date, Checkout Date, Number of Guests.

Billing Information: Room Charges, Additional Expenses.

Process Logic:

Login Module:

The user enters their username and password.

The system verifies the user's credentials.

The user is redirected to the dashboard.

Room Management Module:

Hotel managers can add new rooms to the system. Hotel managers can update room information such as availability and room price.

Hotel managers can delete rooms from the system.

Reservation Module:

Guests can search for available rooms based on their requirements.

Guests can book rooms online.

Hotel managers can view and manage room reservations.

Guest Management Module:

Hotel managers can add new guests to the system.

Hotel managers can update guest information.

Hotel managers can view guest booking history.

Billing Module:

Hotel managers can generate bills for guests. The system will calculate room charges and additional expenses.

Guests can view and pay their bills online.

Reporting Module:

Hotel managers can generate various reports based on different criteria such as room occupancy, revenue, and guest feedback.

Testing Process:

The Hotel Management System will be tested using the following process:

Unit Testing: Each module will be tested individually to ensure that it works as expected.

Integration Testing: All the modules will be tested together to ensure that they work together seamlessly.

User Acceptance Testing: The system will be tested by actual users to ensure that it meets their requirements.

Hardware and software requirement specifications:

Hardware requirement specification:

Processor: AMD PRO A6-7350 R5 and above

RAM: 2GB RAM

HARD DISC: 5GB

Software requirement specification:

Operating system: WINDOWS 7/8/9/10/11

Front end language: HTML, CSS, and JavaScript

Back End Language: PHP and MySQL

Future scope and further enhancement of the project:

The project of a hotel management system using PHP and MySQL database connectivity has a lot of potential for future scope and further enhancement. Some of the possible ways to enhance and extend the project are:

Adding more features: The hotel management system can be enhanced by adding more features to it, such as room service, housekeeping, laundry service, and billing management. This would make the system more comprehensive and useful for hotel staff.

Integrating with third-party services: The hotel management system can be integrated with third-party services such as online booking platforms and payment gateways to streamline the reservation and payment process.

Implementing machine learning algorithms: Machine learning algorithms can be used to analyse customer data and provide insights into customer behaviour and preferences. This can help hotel staff to provide a more personalized and efficient service to their customers.

Mobile app development: The hotel management system can be extended to mobile devices by developing a mobile app that allows guests to book rooms, order services, and check their bill.

Bibliography

The following resources were referred to during the development of the Hotel Booking Website. These include official documentation, tutorials, and online materials that guided the implementation of various features and technologies:

- React.js Documentation: Comprehensive guide to building the front-end interface.
- Node.js Documentation: Resource for managing server-side logic and API integrations.
- MongoDB Documentation: Helped in understanding database structure, queries, and deployment via MongoDB Atlas.
- Online Tutorials: Various tutorials from platforms like W3Schools, Freedcamp, and YouTube for practical insights.
- Articles and Blogs: Blogs on software development trends, best practices in full stack development, and deployment strategies.

References

- React.js Documentation: <https://react.dev/>
- Node.js Documentation: <https://nodejs.org/>
- MongoDB Documentation: <https://www.mongodb.com/docs/>
- Additional Resources: Tutorials and technical support forums like Stack Overflow and developer blogs.